

The Local Compressor

Micro View - Causal Base

Input Video Segment S_i



Vision Encoder
(e.g., SigLIP)

Small Vision-Language Model
(SVLM)

System
Prompt

Visual Tokens
 $\mathbf{X}_i$

User Query
 Q

Memory Tokens
 $\mathbf{M} (k_{max}=128)$

Intercept $\mathbf{h}_i^{rel}$

Causal Attention: $\mathbf{M}$
attends to all conditions

Full Memory
Tokens (k_{max})



Inference-Only Bypass

Training-Free ATA

Zero-shot Relevance

$$s_i = \sigma \left((\mathbf{w}_{\text{yes}} - \mathbf{w}_{\text{no}})^\top \mathbf{h}_i^{\text{rel}} \right) \in (0, 1)$$

ATA
Controller

Global Budget
 B_{max}

Segment 1: High Score s_i



Keep all details ($k_i = 128$)

Segment 2: Low Score s_i



Keep temporal anchors ($k_i = 4$)

The Global Decoder

Macro View - Time-Aware Sequence

<t=0.0s>



<t=2.0s>

<t=4.0s>

User Query
 Q

Large Language Model (e.g., Qwen3-4B)

Global Self-Attention

...

Global Self-Attention

Final Answer